

CoE 1195 Senior Project

ChessConnect

Final Design

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Department of Electrical and Computer Engineering



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Submitted to Dr. Laura Wieserman

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Final Design – Bill of Materials

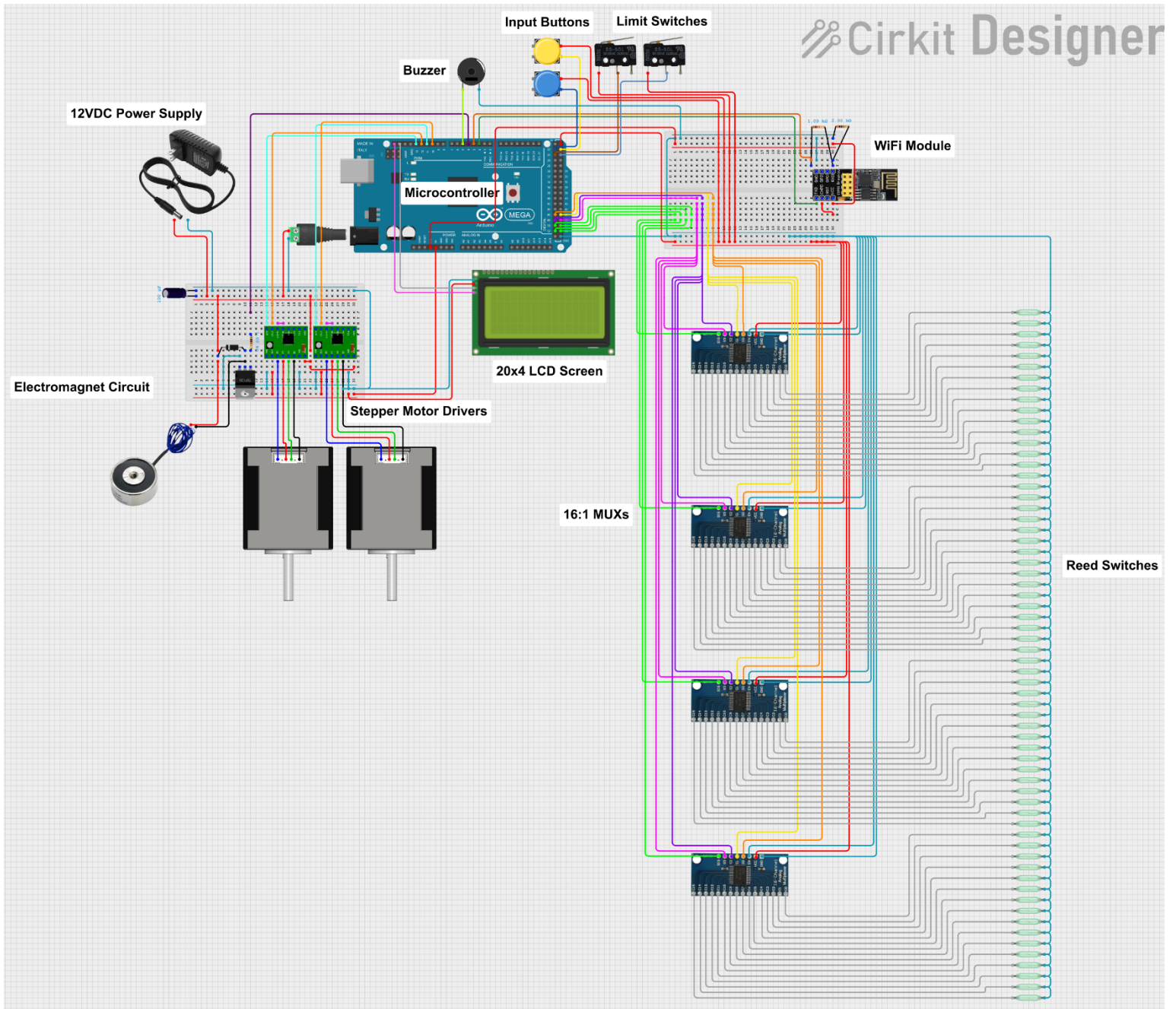
Product Number	Product Name	Amount	Cost Per Unit	Description	Manufacturer	Standards	Cost
Mega 2560 R3	ELEGOO Mega R3 (Atmega 2560)	1	\$18.39	Microcontroller for the system, handling core functions and communication across components	ELEGOO	RoHS Compliant	\$18.39
3-01-0254	ESP8266 (ESP-01) Serial WiFi Transceiver (3pcs)	1	\$8.99	WiFi module, used to connect with Lichess	HiLetgo	IEEE 802.11 b/g/n	\$8.99
ESP8266 ESP-01	ESP8266 (ESP-01) Breakout Board (5pcs)	1	\$7.99	Provides easier connection to the WiFi module	DIYmall	IEEE 802.11 b/g/n	\$7.99
17HS13-0404S1	STEPPERONLINE Nema 17 Stepper Motor	2	\$13.99	Motors used for the XY rail system, facilitates movement of chess pieces	STEPPERONLINE	RoHS, NEMA 17, ISO, CE	\$27.98
A4988	HiLetgo A4988 Stepstick Stepper Motor Driver Module (5pcs)	1	\$10.19	Motor drivers used to control the motors	HiLetgo	N/A	\$10.19
CD74HC4067	16-Channel Analog/Digital Multiplexer (10pcs)	1	\$9.99	Multiplexers used to reduce pins needed for the 64 reed switches	ACEIRMC	IEC 60134	\$9.99
BJ-TIP-120	TIP-120 Darlington NPN Transistor (20pcs)	1	\$7.19	Used to control the electromagnet	BOJACK	N/A	\$7.19
3-01-0103-B	HiLetgo CP2102 USB to TTL Module Serial Converter	1	\$7.39	Used to flash firmware of WiFi module	HiLetgo	RoHS Compliant	\$7.39
E325	Normally Open Reed Switch (2mm x 14mm) (30pcs)	3	\$8.99	Reed Switches used to detect magnetic contact of a chess piece on a given square	Gebildet	IP54, X-10	\$26.97
TS-P25/11	Round 12V Electromagnet (50N, 25 x 11mm)	1	\$10.20	Electromagnet used to move a piece to a new position	TOMSHIELE	IEC 60947-5-1	\$10.20
SFM-27-W	Piezo Electronic Buzzer (3pcs)	1	\$5.49	Electronic Buzzer used to notify user of an issue	WEICHUANGJING	N/A	\$5.49
710525506115	EG STARTS 30mm Arcade Push Button (6pcs)	1	\$12.99	Push Buttons used to navigate UI and control the clock when in a game	EG STARTS	N/A	\$12.99
B07DL13RZH	Breadboard Kit (4pcs)	1	\$8.79	Breadboards used to simplify wiring	REXQualis	EIA RS-310	\$8.79
732214748156	IIC I2C 20x4 LCD Screen Module (3pcs)	1	\$13.59	LCD Screen used to display UI and timer during a game	Hosyond	RoHS Compliant, I2C, SPI, ISO 9001	\$13.59
788125788910	8mm x 3mm Neodymium Magnets (50pcs)	1	\$4.79	Magnets inserted inside of each chess piece to allow for movement using the electromagnet	FINDMAG	ISO 13024, IEC 60404-8-1, EN 71-3	\$4.79
KW12-3	HiLetgo Micro Limit Switch with Roller Lever Arm (10pcs)	1	\$5.99	Limit Switches used for calibration of XY rail system before use to ensure accurate positioning	HiLetgo	IEC 60947, RoHS Compliant	\$5.99
B07N1S412Y	16in x 16in Chess Board Sticker	1	\$14.99	Chess board sticker placed on top of the chassis to show the squares	N/A	N/A	\$14.99
150-LP	500mm V-Slot 20x20 Linear Rail	5	\$4.79	Linear Rails used in the XY rail system	Openbuilds	N/A	\$23.95
TBL6MM205	GT2 20 teeth Pulley - 5mm bore (5pcs)	1	\$6.99	Teethed pulleys used in the XY rail system	WINSINN	ISO 5296	\$6.99

B0B1ZJ342W	Smooth GT2 Idler Pulleys Without Teeth - 5mm bore (8pcs)	1	\$11.99	Smooth pulleys used in the XY rail system	DODOBEI	ISO 5296	\$11.99
MOTO5mbelt	5m GT2 Timing Belt - Width 6mm	1	\$11.99	Timing belt used in the XY rail system	MOTOU	ISO 5296, ASTM D2000	\$11.99
BE063-2	Small Pulley V-Shape Wheel 6mm bore (10pcs)	1	\$12.99	Wheels used to move parts in the XY rail system	befenybay	N/A	\$12.99
B0D2L46TTZ	Aluminum Spacer 5mm Bore x 10mm OD x 6mm Length (30pcs)	1	\$5.99	Spacers used to create gaps between wheels and beams	uxcell	N/A	\$5.99
B-3Dluomu-10.25	Eccentric Spacer 5mm Bore (20pcs)	1	\$8.99	Spacers used to create gaps between wheels and beams	DIAEMI	N/A	\$8.99
170	Mini V Wheel Precision Shim - 5mm ID x 8mm OD x 1mm Thick	8	\$0.22	Shims used for fine adjustment, allowing for smooth movement	Openbuilds	N/A	\$1.76
B0CRL3GJM2	.25 kg 3D Printing Filament	1	\$4.88	3D printing filament used to create chess pieces and carriages for the XY rail system	Likesilk	ISO/ASTM 52900	\$4.88
202093874	3/16 in. x 2 ft. x 4 ft. Underlayment Project Panel	1	\$13.34	3/16 inch wood used as the roof of the casing, thinner than 1/2 inch wood used for the walls to allow for better magnetic contact	Home Depot	ANSI/HPVA HP-1, APA PRP-108	\$13.34
313354418	1/2 in. x 2 ft. x 4 ft. Radiata Pine Plywood Project Panel	2	\$23.94	1/2 inch wood used to construct the casing for internal components and XY rail system	Home Depot	ANSI/HPVA HP-1, APA PRP-108	\$47.88
A-0001	M2 M3 M4 M5 Socket Head Cap Screws Nuts Assortment Kit	1	\$16.89	Assorted screws used to assemble XY rail system	ZTF	ISO 4762	\$16.89
AEP2EA-A5	12 Volt 2A Power Adapter Supply	1	\$7.99	12V DC Power supply used to power the microcontroller	TMEZON	IEC 62368-1, IEEE C62.41	\$7.99
DCPLUG-MF	DC Power Connector 5.5mm x 2.1mm	1	\$5.59	Barrel connectors used to interface the power supply with other components	CENTROPOWER	EIA-364, IEC 60320	\$5.59
Total:							\$383.13

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Final Design – Circuit Diagram



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Final Design – Design Schematics

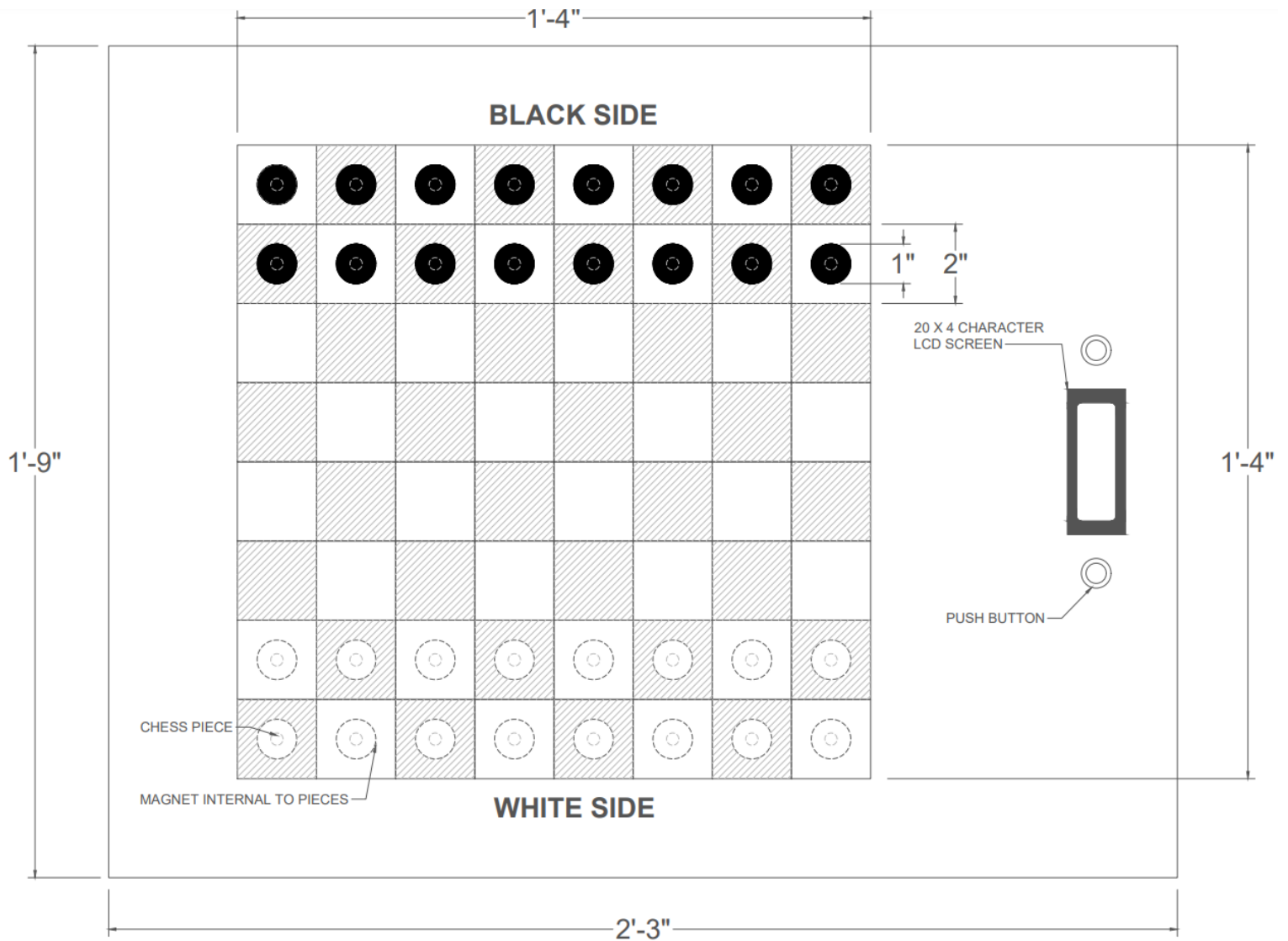


Fig. 1: Overall view of ChessConnect

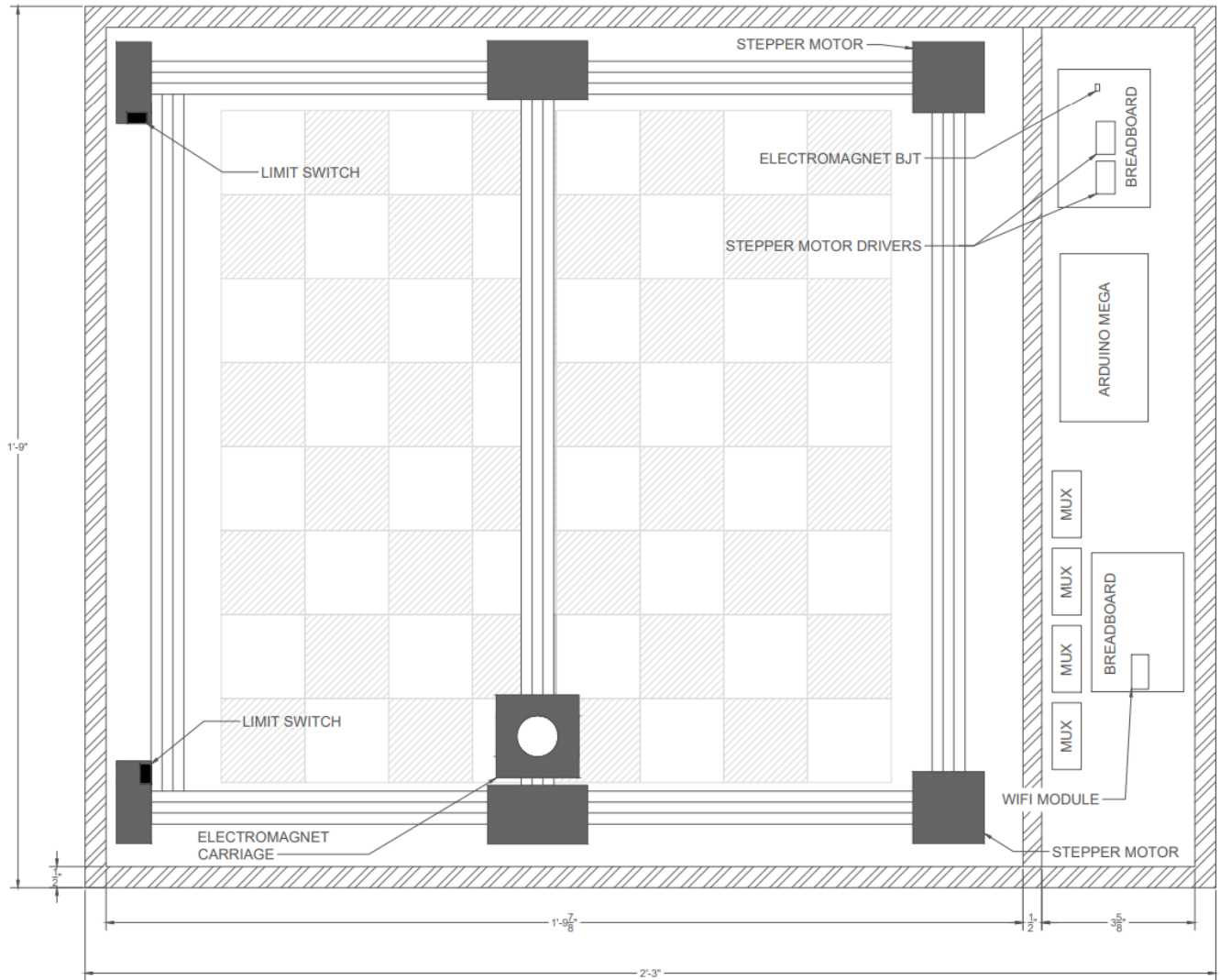


Fig. 2: Internal view of ChessConnect

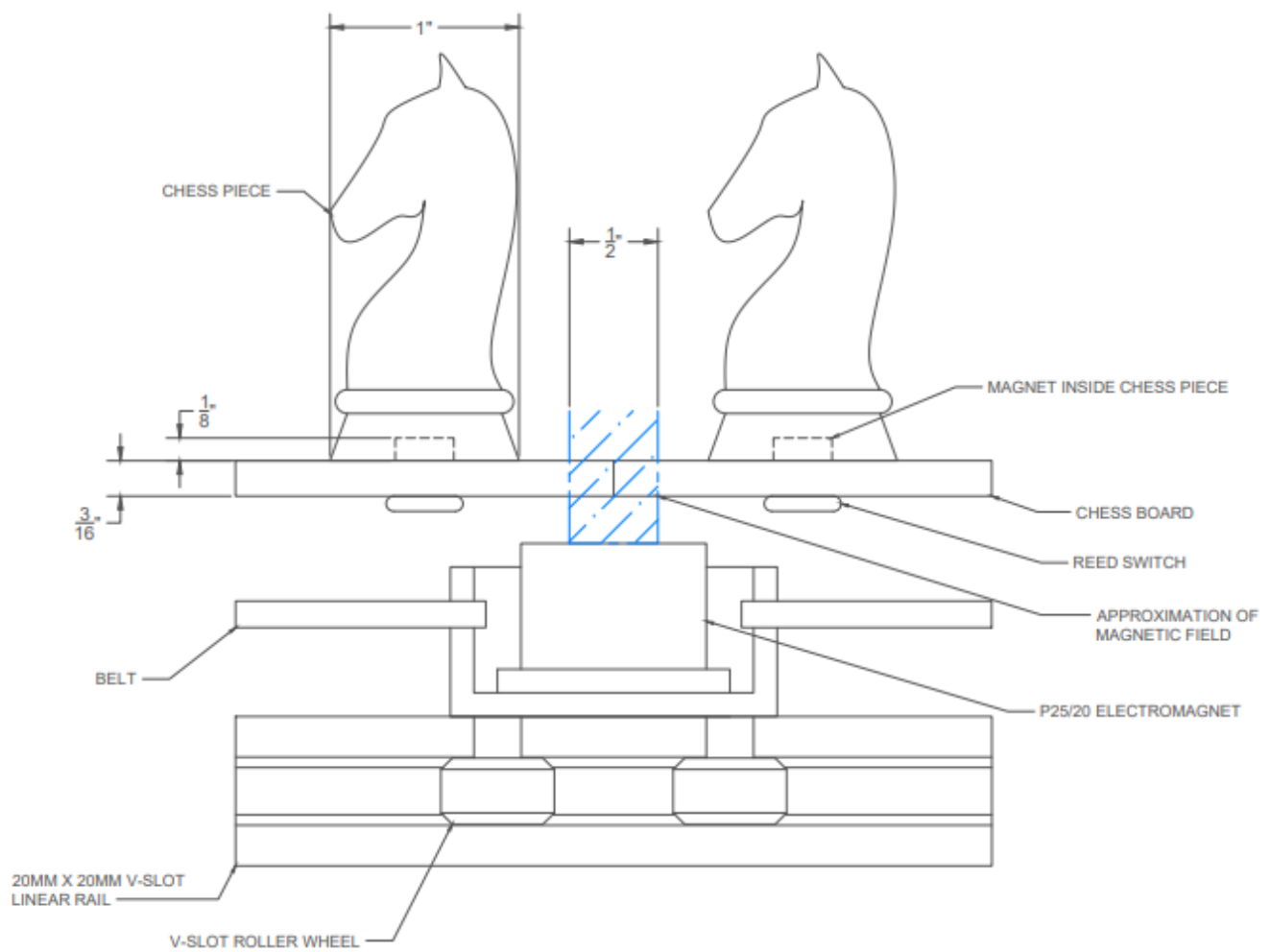


Fig. 3: Enlarged side view of ChessConnect

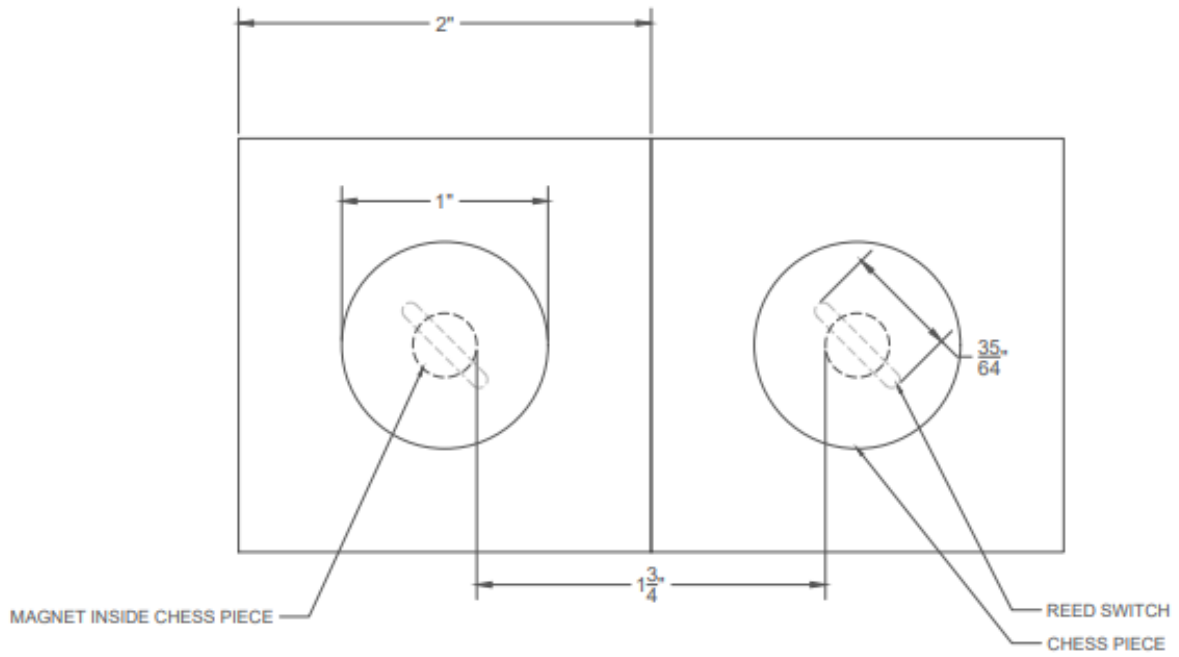


Fig. 4: Enlarged top view of ChessConnect

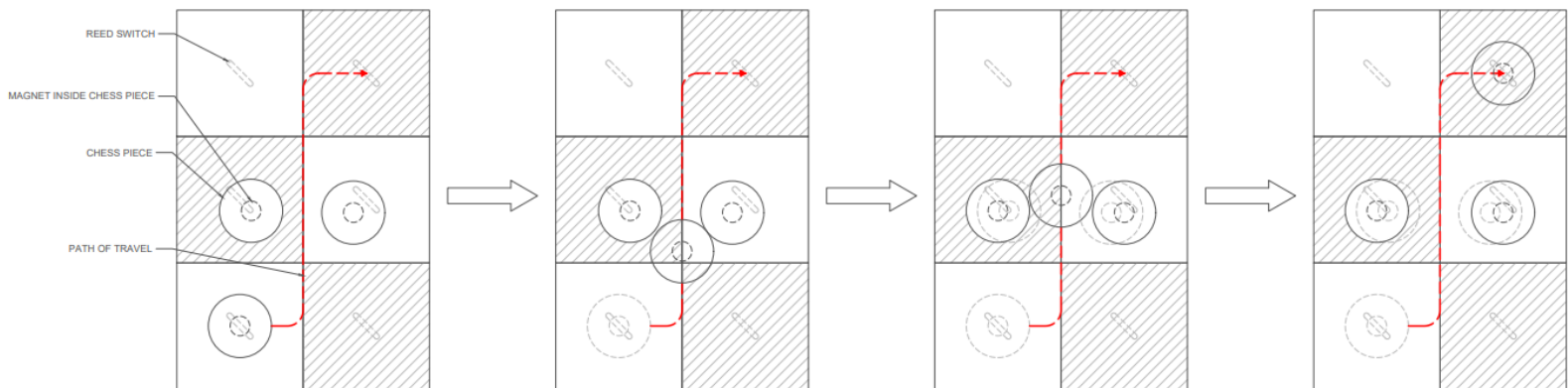


Fig. 5: Piece movement path

